

ChatGPT - development

give from start where you give the create environment
.venv all command i need that installation of all librarys
and packeges also

Here's the complete Mac + VS Code + TensorFlow AircraftWingClassifier setup from scratch.

1. Create Project Folder

Open Terminal:

```
</> Bash
```

```
mkdir
```

```
cd
```

Verify:

```
</> Bash
```

Expected:

2. Create Virtual Environment

Check Python:

```
</> Bash
```

```
--version
```

Expected:

Create environment:

```
</> Bash
```

```
-m
```

Verify:

```
</> Bash
```

```
ls
```

Expected:

3. Activate Environment

```
</> Bash
```

```
source
```

Expected:

4. Upgrade pip

```
</> Bash
```

```
-m
```

```
--upgrade
```

Verify:

```
</> Bash
```

```
--version
```

5. Install Required Libraries

```
</> Bash
```

```
</> Bash
```

```
</> Bash
```

```
</> Bash
```

or install all together:

```
</> Bash
```

6. Verify Installations

TensorFlow:

```
</> Bash
```

```
-c "import tensorflow as tf; print(tf.__version__)"
```

OpenCV:

```
</> Bash
```

```
-c "import cv2; print(cv2.__version__)"
```

NumPy:

```
</> Bash
```

```
-c "import numpy as np; print(np.__version__)"
```

7. Open VS Code

```
</> Bash
```

8. Create Project Structure

Terminal:

```
</> Bash
```

```
mkdir  
mkdir  
mkdir  
  
mkdir  
  
touch  
touch  
touch  
touch  
touch
```

9. Create Counter File

```
</> Bash
```

```
echo 0
```

Verify:

```
</> Bash
```

```
cat
```

Output:

10. Check Folder Structure

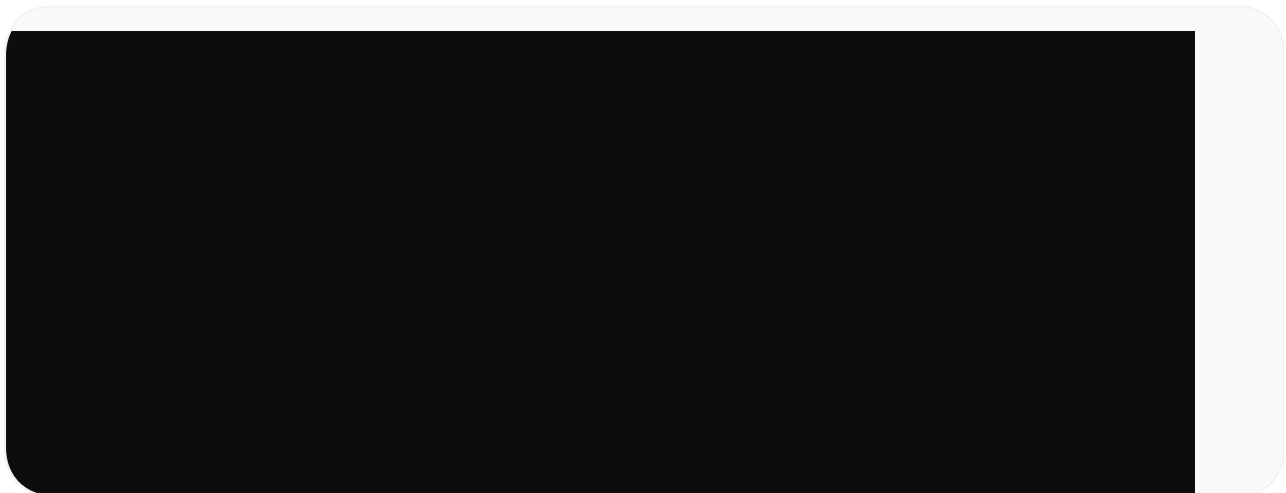
```
</> Bash
```

```
find
```

Should look similar to:

11. Add Your Images

Copy images:



Count images:

```
</> Bash
ls          wc -l
```

```
</> Bash
ls          wc -l
```

12. Save Installed Packages

```
</> Bash
```

Verify:

```
</> Bash
```

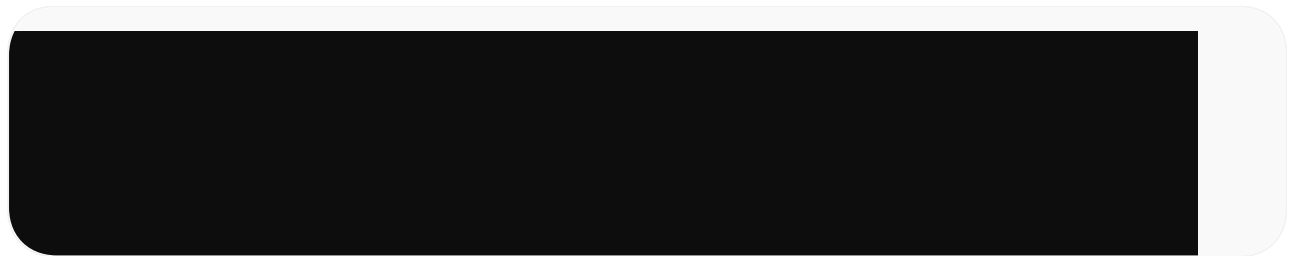
```
cat
```

13. Train Model

After pasting the `train.py` code:

```
</> Bash
```

Expected:

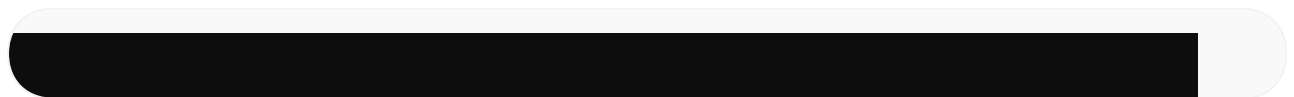


14. Verify Model

```
</> Bash
```

```
ls
```

Expected:



15. Test Single Image

```
</> Bash
```

16. Run Smart Camera

```
</> Bash
```

Controls:

17. Daily Startup Commands

Every time you open the project:

```
</> Bash
```

```
cd  
  
source
```

18. If You Clone or Move the Project

Install everything again from requirements:

```
</> Bash
```

```
source  
  
-r
```

19. Useful Commands

Count dataset:

```
</> Bash
```

```
echo "LHS:"  
ls                wc -l  
  
echo "RHS:"  
ls                wc -l
```

Check TensorFlow:

</> Bash

```
-c "import tensorflow as tf; print(tf.config.list_physical_devices())"
```

Check model exists:

</> Bash

```
ls
```

Retrain manually:

</> Bash

Run camera:

</> Bash

This is the complete setup from zero to a self-improving LHS/RHS classifier on your MacBook Air.